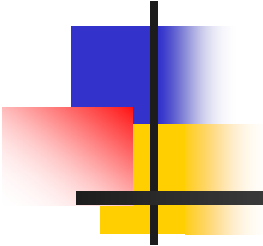


Advanced Manufacturing  
Technology & Prototyping  
MDP494\_UG:2018



# **Lecture 5\_Part 1**

## **Abrasive Jet Machining**

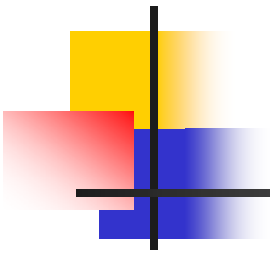
**DR. Moustafa Mahmoud**

Ph. D. Eng.

Design & Production Dept.,

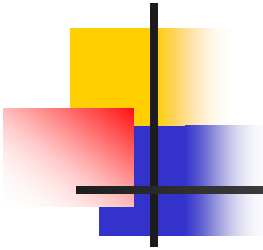
Faculty of Engineering, Ain Shams University

[m\\_m\\_sayed@eng.asu.edu.eg](mailto:m_m_sayed@eng.asu.edu.eg)



# **ABRASIVE JET MACHINING**

## **(AJM)**



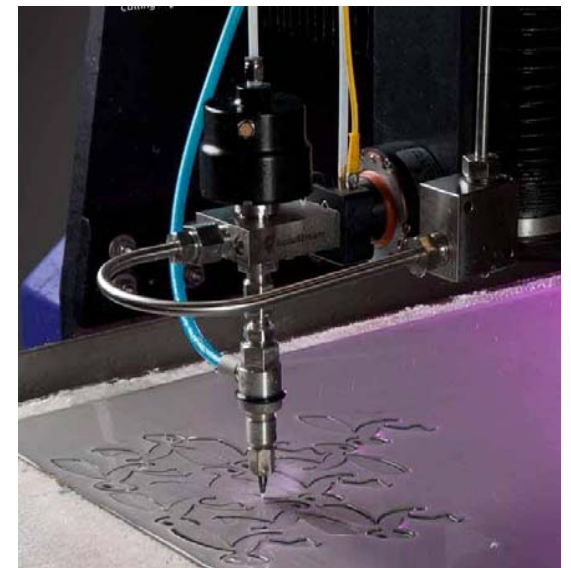
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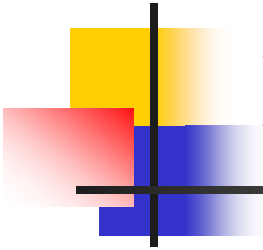
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- **Introduction to AJM**
- **Schematic diagram**
- **Principle & Working**
- **Mechanism**
- **AJM system & subsystem**
- **Process parameters & effect**
- **Process Capability**
- **Application**
- **Advantages & disadvantages**
- **Products**

# Introduction to AJM

- Abrasive Jet Machining it is also known as Abrasive Micro Blasting, Pencil Blasting and Micro Abrasive Blasting.
- It is an abrasive blasting process that uses abrasives propelled by high velocity gas to remove material from the work piece.

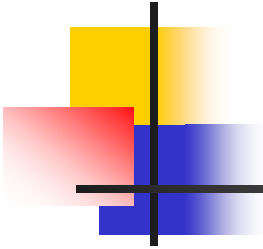




# Definition

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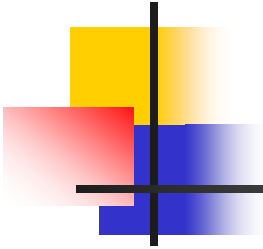
- It is the material removal process where the material is removed by high velocity stream of air/gas and abrasive mixture.
- A focused stream of the abrasive particles, carried by high pressure air or gas is made to impinge on the work surface through a nozzle and work material is removed by erosion by high velocity abrasive particles.



# Process Principle

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- Abrasive particles are made to impinge on work material at high velocity.
- Abrasive particles is carried out by carrier gas/air.
- High velocity stream of the abrasives is generated by converting pressure energy of carrier gas or air to its kinetic energy and hence high velocity jet.

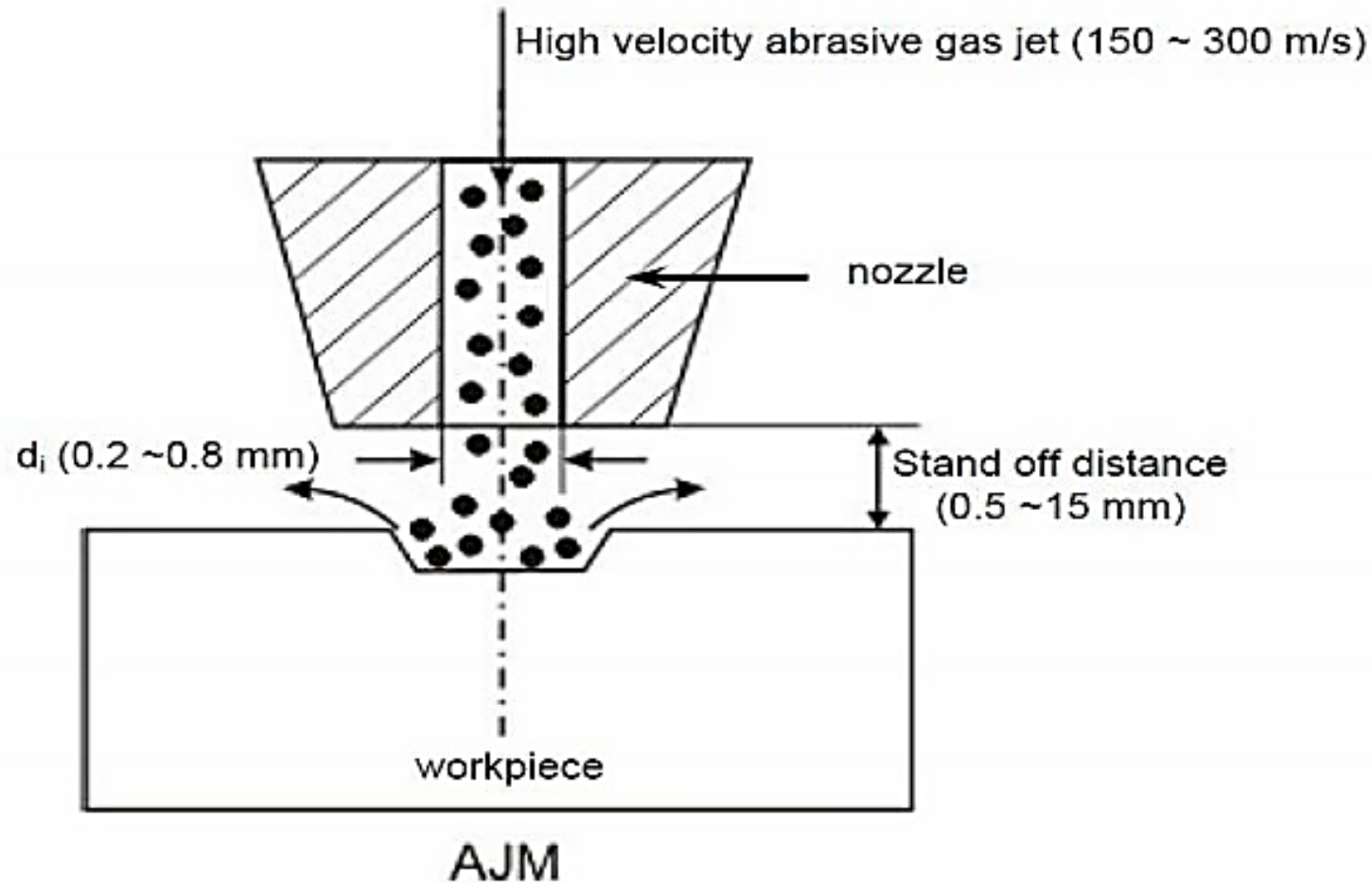


## Process Principle, cont.

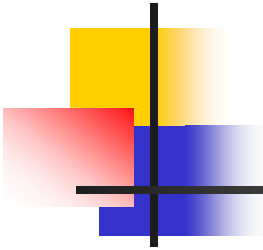
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- Nozzles directs the abrasive jet in a controlled manner onto workpiece.
- Metal cutting action by micro-cutting as well as the brittle fracture of the work material.
- Different from conventional sand blasting/shot blasting, finer abrasive grits are used and process or machining parameters are easily controllable.

# Schematic diagram







# Mechanism

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- 1. Fine particles (0.025mm) are accelerated in a gas stream.
- 2. The particles are directed towards the focus of machining.
- 3. As the particles impact the surface, it causes a micro fracture, and gas carries fractured particles away.

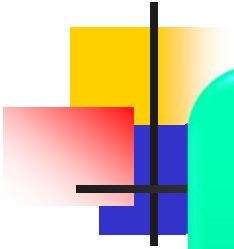
# Mechanism, cont.

An abrasive is a material often a mineral that is used to shape or finish a work piece through rubbing which leads to part of the work piece being worn away by friction.

## **Abrasives used:**

1. Aluminium oxide
2. Silicon carbide
3. Sodium bicarbonate
4. Dolomite
5. Glass beads





# AJM System

**Air  
compressor**

**Air Filter**

**Pressure  
Gauge**

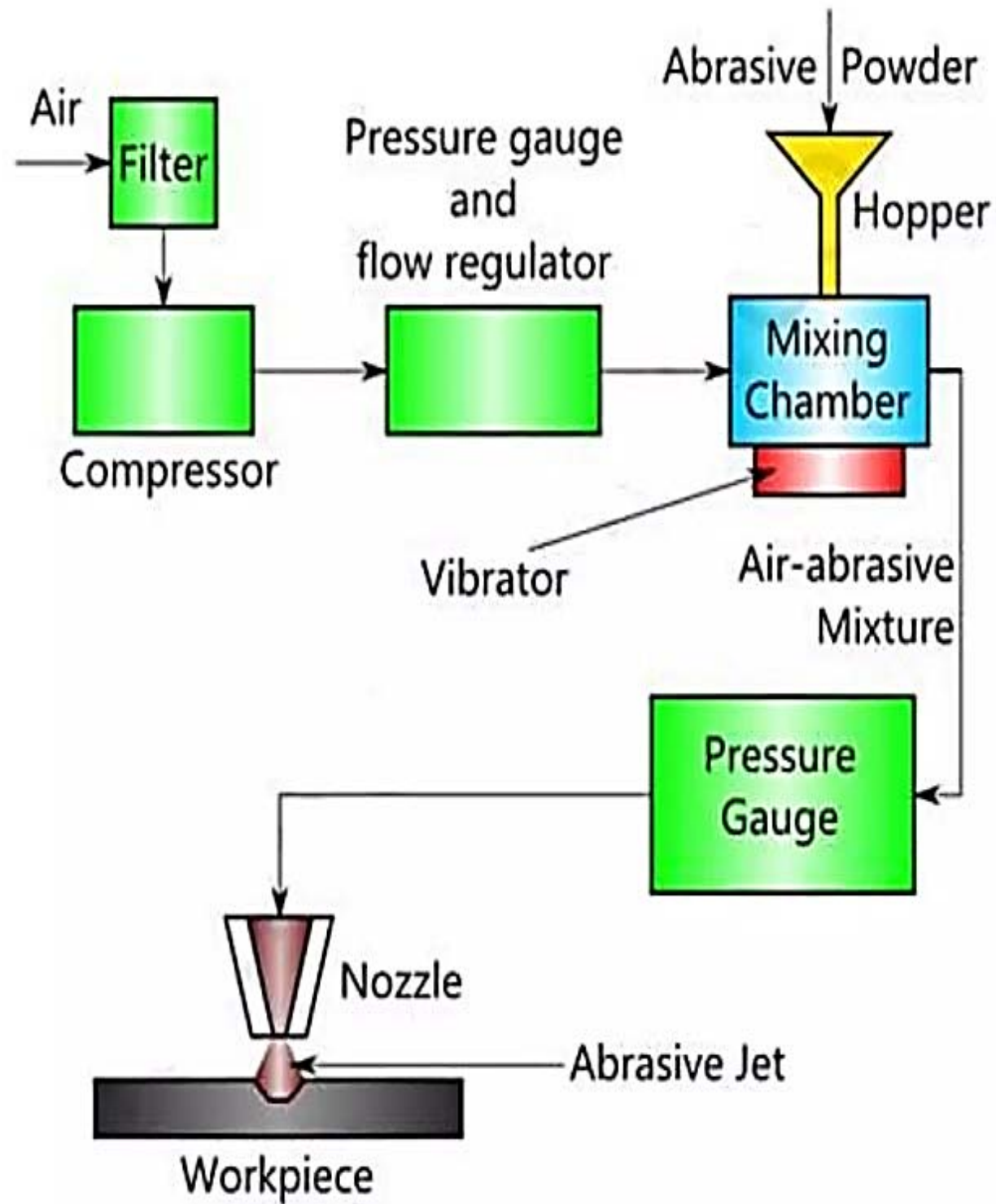
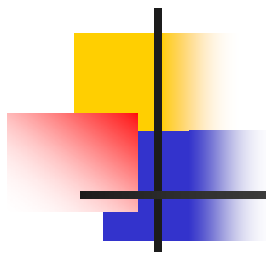
**Flow  
regulator**

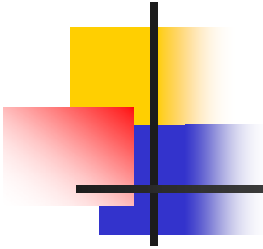
**Mixing  
Chamber**

**Hopper**

**Vibrator**

**Nozzle**





# Process Parameters

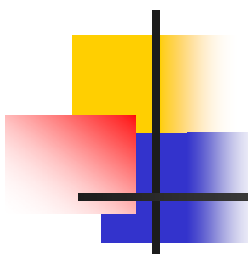
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- **Process output criteria:**

- Material removal rate
- Geometry and surface finish of workpiece.
- Wear rate of nozzle

- **Process criteria are influenced by process parameters:**

- Abrasives: types, rate, size
- Carrier Gas: air or other
- Abrasive jet: pressure & flow rate
- Nozzle: size, stand off



# Process Parameters, cont.

## **Abrasives**

- a) Material –  $\text{Al}_2\text{O}_3$ ; SiC; glass beads.
- b) Shape – irregular/regular
- c) Size – 10 to 50 microns
- d) Mass flow – 2-20 gm/min

## **Carrier Gas**

- a) Composition – Air,  $\text{CO}_2$ ,  $\text{N}_2$
- b) Density –  $1.3 \text{ kg/m}_3$
- c) Velocity - 500 to 700 m/s
- d) Pressure - 2 to 10 bar
- e) Flow rate - 5 to 30 microns

## **Abrasive Jet**

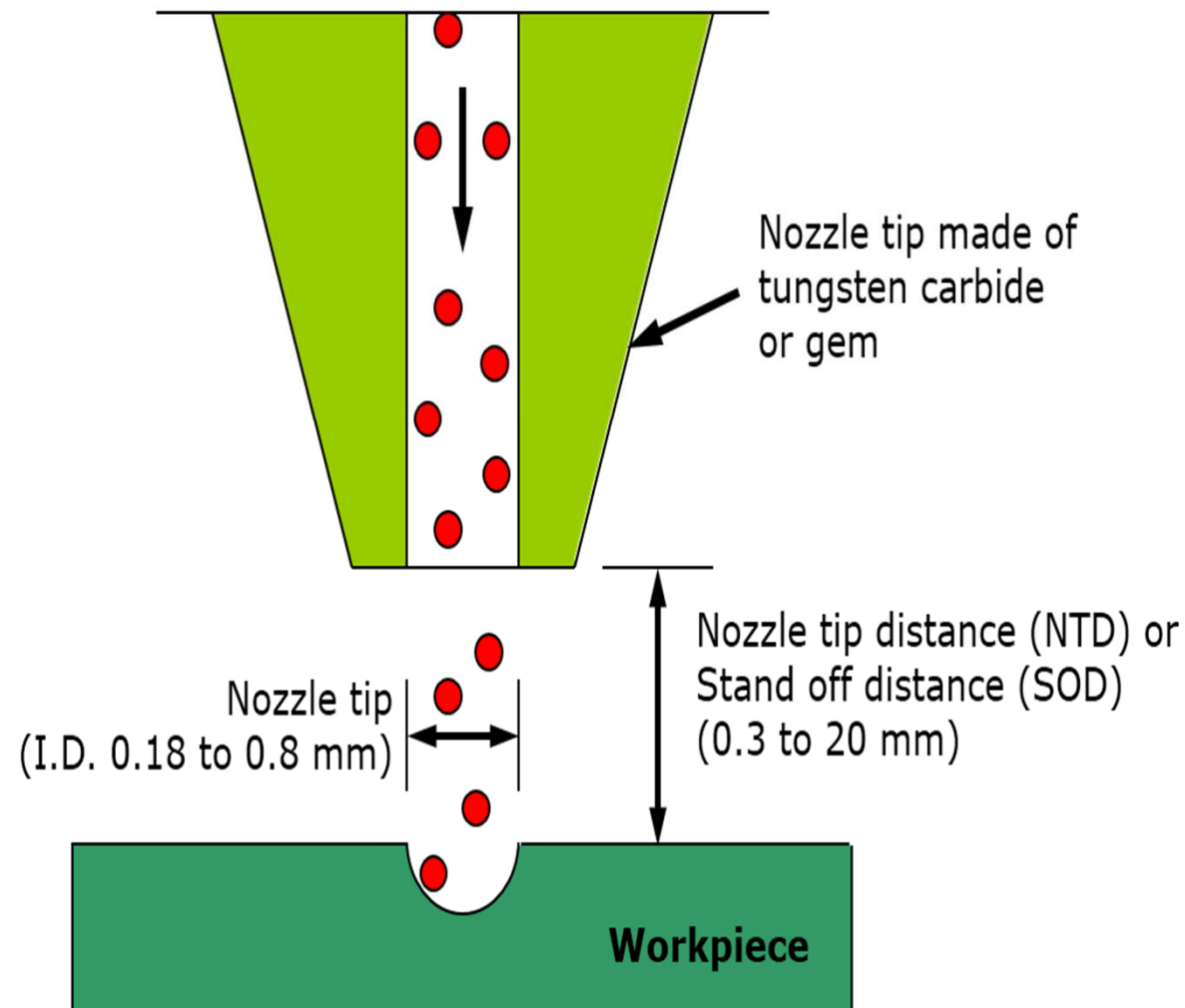
- a) Velocity - 100 to 300 m/s
- b) Stand off distance (SOD)– 0.5 to 15mm.
- c) Impingement angle – 60 to 90 deg.

## **Nozzle**

- a) Material – WC/Sapphire
- b) Diameter – 0.2 to 0.8 mm
- c) Life – 300 hours for sapphire, 20 to 30 hours for WC

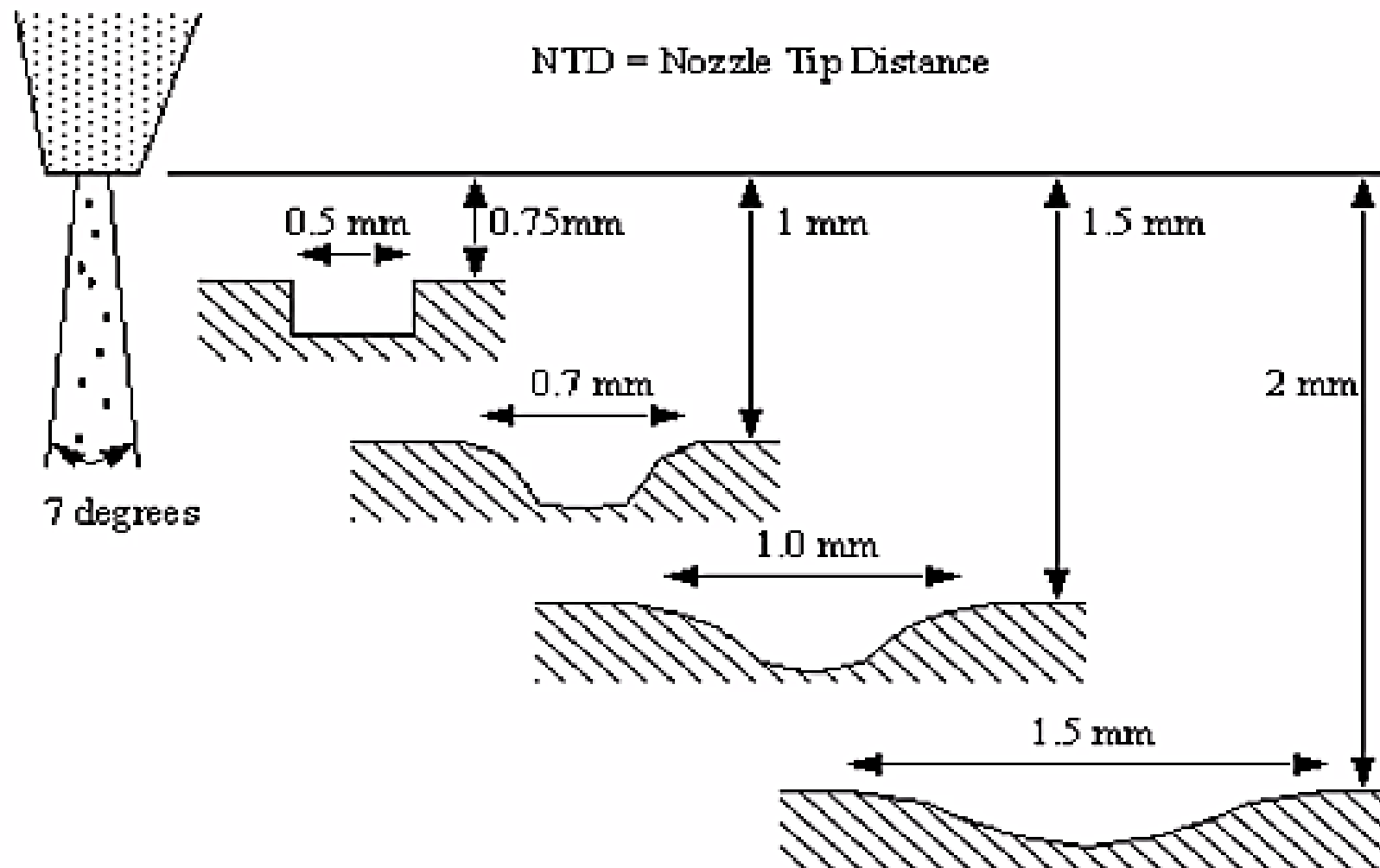
# Process Parameters, cont.

Gas (air) + abrasive particles  
velocity: 150 to 300 m/sec.



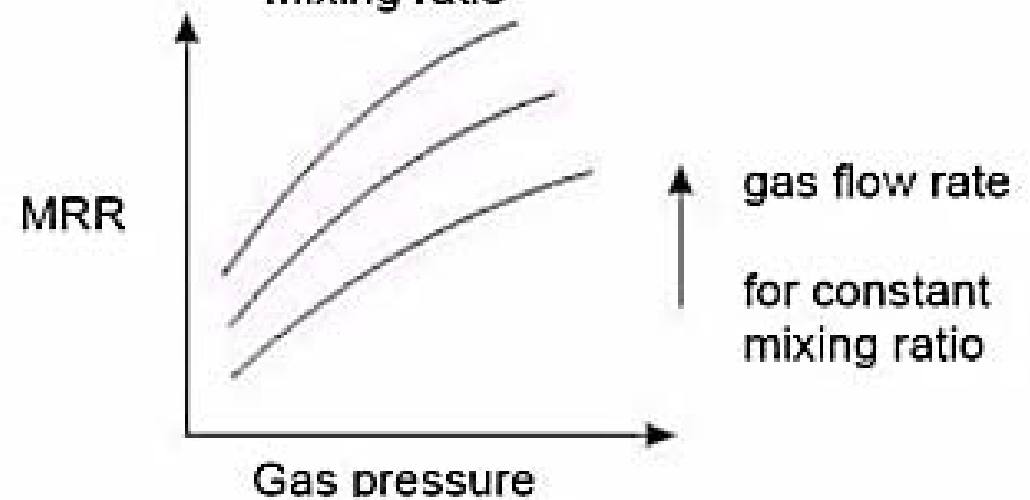
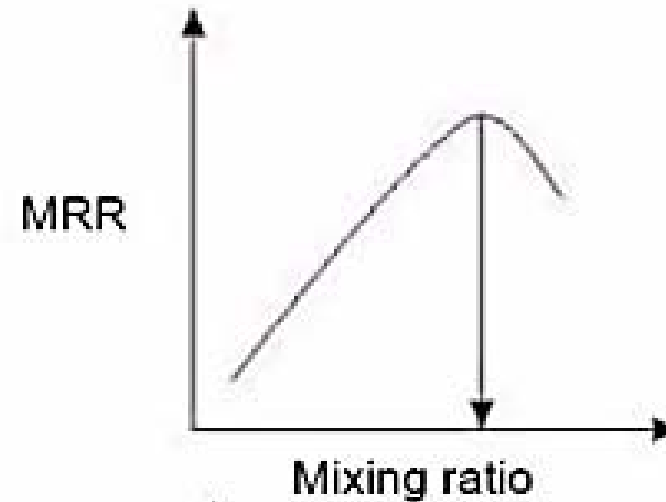
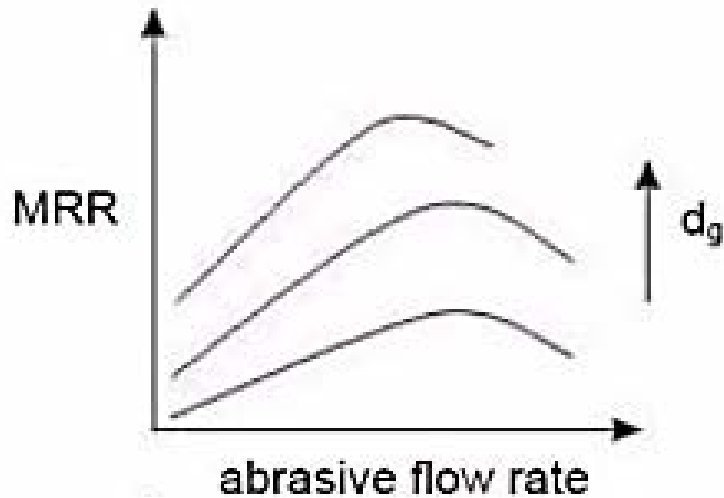
# Process Parameters, cont.

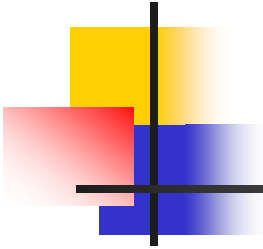
## Stand off distance





# Effect of Process Parameters on MRR





# Material Removal rate

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- The metal removal rate (**MRR**) of AJM can be obtained by the following equation or formula, Given below:

$$MRR = KNd_a^3 v^{3/2} \left( \frac{\rho_a}{12H_w} \right)^{3/4}$$

where  $K$  = constant

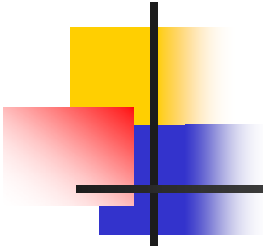
$N$  = number of abrasive particles impacting/unit area

$d_a$  = mean diameter of abrasive particles,  $\mu\text{m}$

$\rho_a$  = density of abrasive particles,  $\text{kg/mm}^3$

$H_w$  = hardness number of the work material

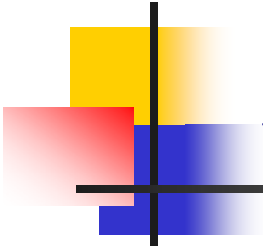
$v$  = speed of abrasive particles,  $\text{m/s}$



# Process Capability

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- Material removal rate:  $0.015 \text{ cm}^3/\text{min}$
- Narrow slots:  $0.12 \text{ to } 0.25\text{mm} \pm 0.12\text{mm}$
- Surface finish:  $0.25 \text{ micron to } 1.25 \text{ micron}$
- Sharp radius up to  $0.2 \text{ mm}$  is possible
- Steel up to  $1.5 \text{ mm}$  ,Glass up to  $6.3 \text{ mm}$  is possible to cut
- Machining of thin sectioned hard and brittle materials is possible.

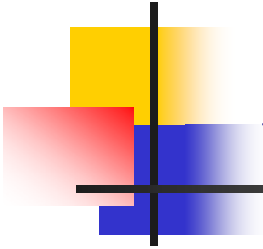


# Applications of AJM

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**The following are some of the operations that can be preformed using abrasive jet machining :**

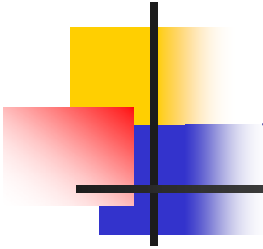
- Drilling
- Boring
- Milling
- Cleaning
- Trimming
- Surface finishing
- Cutting
- Deburring



## Applications, cont.

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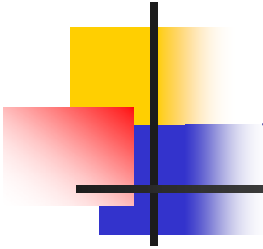
- This is used for abrading and frosting glass more economically as compared to etching or grinding
- Cleaning of metallic smears on ceramics, oxides on metals, resistive coating etc.
- AJM is useful in manufacture of electronic devices , drilling of glass wafers, de burring of plastics, making of nylon and Teflon parts permanent marking on rubber stencils, cutting titanium foils
- Deflashing small castings, engraving registration numbers on toughened glass used for car windows



## Applications, cont.

---

- Used for cutting thin fragile components like germanium, silicon etc.
- Register teaming can be done very easily and micro module fabrication for electrical contact , semiconductor processing can also be done effectively.
- Used for drilling , cutting , deburring etching and polishing of hard and brittle materials.
- Most suitable for machining brittle and heat sensitive materials like glass, quartz, sapphire , mica , ceramics germanium , silicon and gallium



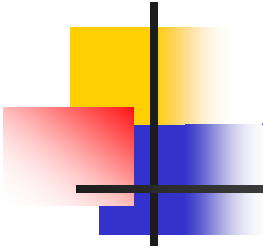
# Advantages of AJM

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- High surface finish can be obtained depending upon the grain sizes:

| Particle size ( in microns) | Surface roughness ( in microns) |
|-----------------------------|---------------------------------|
| 10                          | 0.152 to 0.203                  |
| 25 to 27                    | 0.355 to 0.675                  |
| 50                          | 0.965 to 1.27                   |

- Depth of damage is low ( around 2.5 microns)
- Hard materials and materials of high strength can be easily machined.
- Surface of the work piece is cleaned automatically prior to cutting.

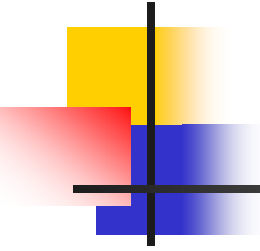


## Advantages of AJM, cont.

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- Capital cost is low and it is easy to operate and maintain AJM.
- Thin sections of hard brittle materials like germanium, mica, silicon, glass and ceramics can be machined.

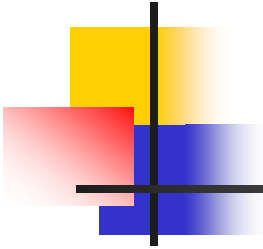




# Limitations of AJM

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- Metal Removal rate is low.
- Nozzle wear is high.
- Nozzle life is limited (300 hours)
- Process tends to pollute the environment so dust collecting system is required.



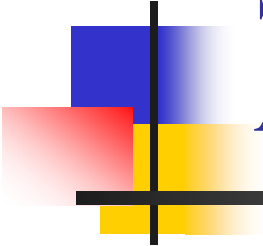
## Limitations of AJM, cont.

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- Limited capacity due to low MRR. MRR for glass is 40 gm/minute.
- Abrasives may get embedded in the work surface, especially while machining soft material like elastomers or soft plastics.
- A dust collection system is a basic requirement to prevent atmospheric pollution and health hazards.

# Products

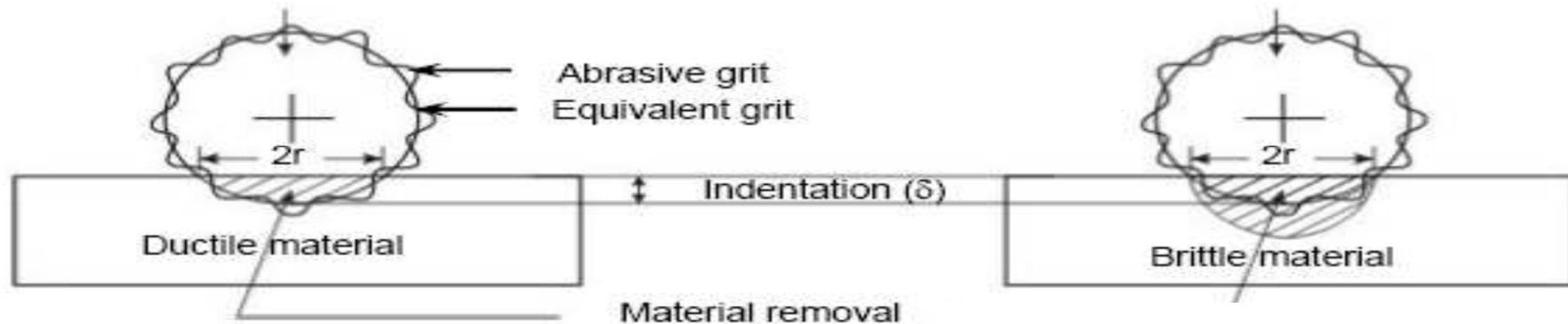




# Solved Examples for calculating MRR of the Abrasive jet machining

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# Material removal models in AJM



For Brittle material

Volume of the material removed

$$\Gamma_B = \frac{1}{2} \left[ \frac{4}{3} \pi r^3 \right] = \left[ \frac{2}{3} \pi (d_g \delta)^{\frac{3}{2}} \right]$$

$$MRR = \left\{ \begin{array}{l} \text{Volume of the material} \\ \text{removed} \\ \text{per grit per cycle.} \end{array} \right\} \times \left\{ \begin{array}{l} \text{Number of} \\ \text{impacts made by} \\ \text{abrasives per second} \end{array} \right\}$$

$$MRR = \Gamma_B \times \left[ \frac{\text{Mass flow rate of abrasives}}{\text{Mass of the abrasive grit}} \right]$$

**For ductile material**

$$\Gamma_D = \frac{1}{2} \left[ \pi (d_g) (\delta)^2 \right]$$

$$MRR = \Gamma_D \times \left[ \frac{\text{Mass flow rate of abrasives}}{\text{Mass of the abrasive grit}} \right]$$

Substituting and simplifying we get.

$$MRR = \left[ \frac{M_a V^2}{2 \sigma_w} \right]$$

$$MRR_{\text{Brittle}} = \left[ \frac{M_a v}{(\rho_g)^{\frac{1}{4}} (\sigma_w)^{\frac{3}{4}}} \right]$$

$\Gamma_D$  = Volume of the material removed

$d_g$  = diameter of abrasive particle

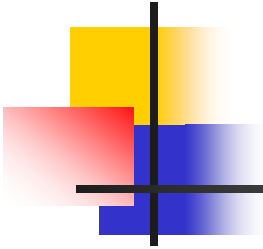
Indentation ( $\delta$ )

Flow strength of material =  $\sigma_w$

$M_a$  Mass flow rate of abrasives

Velocity =  $V$

$\rho_g$  = density of the grit



## Example 1

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Estimate the material removal rate in AJM of a brittle material with flow strength of 4 GPa. The abrasive flow rate is 2 gm/min, velocity is 200 m/s and density of the abrasive is 3 gm/cc.

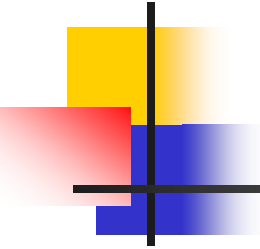
### Data Given:

Flow strength of work material =  $\sigma_w = 4 \times 10^9 \text{ N/m}^2 = 4000 \text{ N/mm}^2$

Abrasive grain density =  $3 \text{ gm/CC} = 3 \times 10^{-3} \times 10^{-3} = 3 \times 10^{-6} \text{ kg/mm}^3$

Mass flow rate of abrasives =  $2 \text{ gm/min} = 2 \times 10^{-3} / 60 \text{ kg/sec}$

Velocity =  $V = 200 \times 1000 \text{ mm/sec}$



## Example 1, cont.

---

$$\text{MRR}_{\text{Brittle}} = \left[ \frac{M_a v}{(\rho_g)^{\frac{1}{4}} (\sigma_w)^{\frac{3}{4}}} \right] = \left[ \frac{2 \times 10^{-3} \times 2000000^{\frac{3}{2}}}{(3 \times 10^{-6})^{\frac{1}{4}} (4 \times 10^9)^{\frac{3}{4}}} \right]$$
$$= 48 \text{ mm}^3 / \text{minutes}$$